

Day 5 — Practice — 5×5 and 7×7

Module: Magic Squares — Bhadrageṇita · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: use the method on multiple odd-order squares.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *siddhāṅka* (modern coinage) (IAST: siddhāṅka (modern coinage); lit. “the common sum of all rows/columns/diagonals”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **5×5 construction.** Place 1 in cell (1, 3) [top-middle]. Apply the stair-step rule. Resulting square (one variant):

17	24	1	8	15
23	5	7	14	16
4	6	13	20	22
10	12	19	21	3
11	18	25	2	9

Magic constant: $5 \times 26 / 2 = 65$. Verify any row, column, or diagonal.

2. **7×7 construction (challenge).** Same algorithm, larger grid. Magic constant: $7 \times 50 / 2 = 175$.
3. **Common mistakes.** Forgetting the “if cell occupied, drop down” rule. Wrong wrap direction. Off-by-one starting position.
4. **Source check.** Nārāyaṇa’s text gives this algorithm for any odd n . Modern translations: T. Hayashi, J. Pereira’s edition of Gaṇita-kaumudī.

5. **Independent practice.** Each student builds a 5×5 unaided. Pair-check. Then a 7×7 if time permits.

Activity (15 min)

A scaled version of *Class 3×3 Magic-Square Tournament* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces Practice — 5×5 and 7×7 .

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).